RMA TRACKER DOCUMENTATION LIBRARY

VERSION 2.1.1

VOLUME IV

Installation, Deployment, Maintenance Guide

RMA Tracker

Provides a comprehensive reference for the installation, deployment, configuration, maintenance, backup, recovery, release management, and long-term support of RMA Tracker Version 2.1.1. This guide is intended for system administrators, IT personnel, software developers, and future maintainers responsible for deploying and supporting the application throughout its lifecycle.

Prepared for: Harp Enterprises

Prepared by: Brennen Gabriel

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1. Introduction

This guide provides complete instructions for installing, configuring, deploying, maintaining, upgrading, backing up, restoring, and supporting RMA Tracker Version 2.1.1. It serves as the primary operational reference for system administrators, IT personnel, software developers, and future maintainers.

1.1. Scope

This volume covers installation, configuration, system requirements, deployment, packaging, backup and recovery, maintenance, troubleshooting, release management, and operational best practices.

1.2. Intended Audience

System Administrators, IT Support Personnel, Software Developers, Deployment Engineers, Technical Support Staff, and Future Project Maintainers.

1.3. Relationship to Documentation Library

Volume I defines requirements, Volume II explains architecture, Volume III documents application usage, and Volume IV focuses on installation, deployment, maintenance, and long-term support.

1.4. Document Conventions

Commands, file paths, configuration files, and application components are referenced consistently throughout the guide using standard technical documentation practices.

1.5. Definitions, Acronyms, and Abbreviations

Defines common terms including RMA, SQLite, JDK, Gradle, Deployment, Backup, Restore, Package, and Release.

1.6. References

References include Volumes I–III, Java SE Documentation, Gradle Documentation, SQLite Documentation, and the RMA Tracker source code.

2. System Requirements

This section defines the minimum and recommended hardware, software, operating system, storage, and security requirements necessary to successfully install, deploy, and operate RMA Tracker Version 2.1.1. Verifying that the target environment meets these requirements before installation helps ensure reliable application performance, compatibility, and long-term maintainability. The information presented in this section is intended to assist system administrators, IT personnel, and deployment engineers in preparing workstations and deployment environments while minimizing installation issues and ensuring consistent operation across all supported platforms.

2.1. Overview

Verify that the deployment environment satisfies all minimum requirements before installing the application.

2.2. Supported Operating Systems

Windows 10/11 (64-bit), macOS 13 or later, and modern Linux distributions supporting Java 17+.

2.3. Hardware Requirements

Minimum: Dual-core processor, 8 GB RAM, 500 MB storage. Recommended: Quad-core processor, 16 GB RAM, SSD.

2.4. Software Requirements

Java 17 or newer, embedded SQLite database, and PDF/print support.

2.5. Java Runtime Requirements

Java Runtime Environment (JRE) or Java Development Kit (JDK) version 17 or newer.

2.6. SQLite Requirements

Embedded SQLite database created automatically during first launch.

2.7. User Permissions

Read/write access to application and backup directories. Administrator rights may be needed during installation.

2.8. Network Considerations

The application operates locally; networking is optional for shared storage or update distribution.

2.9. Storage Requirements

Allow sufficient space for the database, backups, exported reports, and future growth.

2.10. Pre-Installation Checklist

Verify operating system compatibility, Java installation, available storage, permissions, backup location, and installation package integrity.

3. Installation

This section provides detailed procedures for installing RMA Tracker Version 2.1.1 on supported operating systems. It explains how to prepare the target environment, install the application, verify a successful installation, and resolve common installation issues. Following these procedures helps ensure a consistent, reliable deployment across Windows, macOS, and Linux.

3.1. Installation Overview

RMA Tracker is distributed as a desktop application packaged for supported operating systems. The installation process creates the application files, initializes the required directory structure, and prepares the application for first-time execution.

3.2. Application Package Contents

The installation package includes the application executable, required Java runtime (when bundled), supporting libraries, configuration resources, documentation, and application icon resources.

3.3. Installing on Windows

Run the installer, accept the license agreement, choose the installation directory, complete the installation wizard, and launch the application. Verify that the application starts without errors and creates the required data folders.

3.4. Installing on macOS

Open the DMG package, drag the application into the Applications folder, approve the application if prompted by Gatekeeper, and launch RMA Tracker. Confirm database initialization and automatic backup folder creation.

3.5. Installing on Linux

Extract or install the application package, ensure Java 17 or newer is available, assign executable permissions if necessary, and start the application from the desktop launcher or terminal.

3.6. First Launch

During first launch, the application verifies its environment, creates the SQLite database if necessary, initializes default folders, and prepares the backup directory.

3.7. Initial Database Creation

If no database exists, RMA Tracker automatically creates the database schema and prepares the application for normal operation.

3.8. Application Folder Structure

The installation creates directories for application data, backups, exported reports, temporary files, and configuration resources.

3.9. Verifying Installation

Confirm the application launches successfully, the database is accessible, reports can be generated, exports function correctly, and automatic backups are enabled.

3.10. Common Installation Issues

Troubleshooting guidance includes missing Java runtimes, insufficient permissions, blocked executables, damaged installation packages, and directory access issues.

4. Configuration

This section describes the configuration settings used by RMA Tracker after installation. It explains how application directories, database locations, backup folders, configuration files, and default settings can be verified and customized to support organizational deployment standards.

4.1. Configuration Overview

After installation, verify that application settings, storage locations, and backup directories are configured correctly.

4.2. Application Directories

Application files, user data, backups, exported reports, and temporary files should be stored in their designated directories.

4.3. Database Location

The embedded SQLite database is stored within the application's data directory and should be included in regular backups.

4.4. Backup Location

Automatic and manual backups should be stored in a secure location with sufficient storage capacity.

4.5. Configuration Files

Configuration files store application preferences and deployment-specific settings. Changes should be documented and tested before production use.

4.6. Default Settings

The default configuration supports immediate operation following installation while allowing future customization.

4.7. Customizing the Environment

Organizations may customize storage paths, backup locations, and deployment settings to meet operational requirements.

4.8. Verifying Configuration

Verify application startup, database connectivity, backup functionality, report generation, printing, and export capabilities before placing the application into production.

5. Building the Application

5.1. Build Overview

RMA Tracker uses Gradle as its build automation system. The build process compiles the Java source code, resolves dependencies, executes automated tests, packages application resources, and generates release artifacts.

5.2. Development Environment

Install Java Development Kit (JDK) 17 or newer, Gradle (or use the Gradle Wrapper), Git, and a supported IDE such as IntelliJ IDEA or Visual Studio Code.

5.3. Required Software

Required software includes JDK 17+, Gradle Wrapper, SQLite libraries, Git, and project dependencies defined in the Gradle build configuration.

5.4. Gradle Project Structure

The project follows a standard Gradle layout with separate source, test, resource, build, and distribution directories to simplify maintenance and future enhancements.

5.5. Building from Source

Clone the repository, open the project, execute the Gradle build task, resolve dependencies, and verify that compilation completes without errors.

5.6. Running Unit Tests

Execute automated unit tests before every release to validate repository classes, business logic, database operations, and application functionality.

5.7. Creating Release Builds

Generate production-ready artifacts after successful compilation and testing. Verify application metadata, icons, version information, and bundled resources.

5.8. Build Verification

Confirm that the application launches correctly, connects to the SQLite database, generates reports, exports data, and performs backups successfully.

5.9. Common Build Errors

Typical issues include missing Java installations, dependency resolution failures, incorrect Gradle configuration, permission problems, and unsupported Java versions.

6. Packaging and Distribution

This section explains how RMA Tracker is packaged and prepared for distribution to end users. It describes supported package formats, operating-system-specific installers, version management, release preparation, and recommended distribution practices to ensure reliable software deployment.

6.1. Packaging Overview

Packaging converts compiled application files into installable distributions suitable for Windows, macOS, and Linux.

6.2. Creating Executable Packages

Use jpackage to generate native installers and application images that include the required runtime environment where appropriate.

6.3. Building Windows Installers

Create Windows executable or MSI installers with application metadata, icons, shortcuts, and uninstall support.

6.4. Building macOS Packages

Generate DMG or PKG installers that integrate with macOS installation standards and application security requirements.

6.5. Building Linux Packages

Create Linux-compatible packages or application images appropriate for the target distribution.

6.6. Application Icons

Verify that application icons are correctly embedded in all supported package formats for a consistent user experience.

6.7. Version Numbering

Follow semantic versioning practices and ensure version numbers remain consistent across application binaries, documentation, and release notes.

6.8. Code Signing (Future)

Future releases may incorporate digital code signing to improve software authenticity and reduce security warnings during installation.

6.9. Distribution Best Practices

Validate packages on clean systems, include release notes and documentation, verify installers before publication, and maintain archived copies of previous releases.

7. Deployment

This section describes the recommended procedures for deploying RMA Tracker Version 2.1.1 into production environments. It covers new installations, application upgrades, database migration, update deployment, rollback planning, deployment verification, and deployment checklists. Following these procedures helps ensure reliable software delivery, minimizes downtime, and provides a consistent deployment process across all supported operating systems.

7.1. Deployment Overview

Deployment prepares RMA Tracker for production by validating requirements, installing the application, configuring storage, verifying backups, and confirming correct operation.

7.2. New Installation Deployment

Verify system requirements, install the application, initialize the SQLite database, configure backup locations, verify permissions, and perform functional validation.

7.3. Upgrading Existing Installations

Create a verified backup before upgrading. Install the new version, preserve existing data, and validate reports, exports, backups, and printing after the upgrade.

7.4. Migrating Databases

Copy the SQLite database and backup files to the new environment, verify permissions and integrity, and confirm successful startup before retiring the previous installation.

7.5. Deploying Updates

Test updates when possible, distribute approved packages, document version changes, and verify successful deployment after installation.

7.6. Rollback Procedures

If deployment problems occur, restore the previous application version and verified database backup before returning the system to production.

7.7. Deployment Verification

Verify startup, database connectivity, RMA management, reporting, exports, printing, status history, and automatic backups after deployment.

7.8. Deployment Checklist

Confirm requirements, installation, configuration, database access, backups, reports, exports, user acceptance testing, documentation updates, and deployment approval.

8. Backup and Recovery

This section describes the procedures and best practices for protecting RMA Tracker data through regular backups and reliable recovery processes. Following these guidelines helps minimize data loss and ensures business continuity in the event of hardware failure, software corruption, or accidental deletion.

8.1. Backup Overview

RMA Tracker stores application data in an embedded SQLite database. Backups should be performed regularly to protect critical business information.

8.2. Automatic Backups

Version 2.1.1 automatically creates database backups when the application closes successfully. Verify that backup files are created and retained.

8.3. Manual Backups

Administrators may create manual backups before upgrades, maintenance activities, or major data changes.

8.4. Backup Storage Locations

Store backups on reliable local or network storage and maintain additional off-site or cloud copies where organizational policies permit.

8.5. Restoring Backups

Restore the most recent verified backup by replacing the application database while the application is closed.

8.6. Recovery Procedures

Validate recovered data, verify application startup, and confirm reporting, exports, and search functionality after recovery.

8.7. Disaster Recovery

Maintain documented recovery procedures and periodically test backup integrity to reduce downtime.

8.8. Backup Verification

Regularly confirm that backup files are complete, readable, and restorable.

8.9. Recovery Testing

Schedule periodic recovery exercises to ensure restoration procedures remain effective.

8.10. Best Practices

Follow a documented backup schedule, retain multiple backup generations, and protect backup media from unauthorized access.

9. Maintenance

This section outlines the routine maintenance activities required to keep RMA Tracker operating efficiently throughout its lifecycle. Regular maintenance improves application stability, data integrity, and long-term reliability.

9.1. Maintenance Overview

Routine maintenance includes monitoring database health, verifying backups, applying updates, and reviewing storage utilization.

9.2. Database Maintenance

Periodically verify database integrity and archive obsolete information when appropriate.

9.3. Cleaning Temporary Files

Remove unnecessary temporary files and obsolete exports to conserve storage.

9.4. Updating the Application

Install approved application updates after creating a verified backup.

9.5. Monitoring Database Growth

Monitor database size and available storage to ensure adequate capacity.

9.6. Preventive Maintenance

Perform scheduled inspections of application functionality, backups, and reporting.

9.7. Performance Monitoring

Observe application startup time, search performance, report generation, and backup duration.

9.8. Maintenance Schedule

Establish daily, weekly, monthly, and annual maintenance tasks appropriate for the deployment.

9.9. Health Checks

Verify successful startup, database connectivity, backup completion, printing, exports, and report generation.

10.Troubleshooting

This section provides guidance for diagnosing and resolving common issues encountered during installation, deployment, maintenance, and everyday operation of RMA Tracker. Systematic troubleshooting reduces downtime and supports rapid recovery.

10.1. Startup Issues

Check Java installation, application permissions, and database accessibility if the application fails to start.

10.2. Installation Failures

Verify installer integrity, operating system compatibility, and administrator privileges.

10.3. Database Problems

Inspect the SQLite database for corruption, restore from backup if required, and verify file permissions.

10.4. Backup Failures

Confirm backup directory permissions, available storage, and successful completion of automatic backup routines.

10.5. Restore Problems

Ensure the application is closed before replacing the database and validate the selected backup.

10.6. Build Errors

Review Gradle output, dependency configuration, and Java version compatibility.

10.7. Deployment Problems

Confirm all required files were packaged and deployed correctly across supported environments.

10.8. Performance Issues

Investigate hardware resources, database size, and unnecessary background processes.

10.9. Log File Analysis

Review application and build logs to identify warnings, exceptions, and error messages.

10.10. Support Procedures

Document issues, collect diagnostic information, and follow organizational support and escalation procedures.

11.Developer Release Process

This section documents the standardized release process used to prepare, validate, package, and publish new versions of RMA Tracker. A consistent release process helps ensure software quality, repeatable deployments, and complete release documentation.

11.1. Release Overview

Each release should follow a documented workflow including development completion, testing, packaging, verification, and publication.

11.2. Version Numbering

Use consistent semantic versioning across the application, documentation, installers, and release notes.

11.3. Pre-Release Testing

Execute unit, integration, and manual regression testing before approving a release.

11.4. Build Verification

Confirm successful compilation, dependency resolution, packaging, and application startup.

11.5. Packaging Checklist

Verify icons, version information, installers, bundled runtime, documentation, and license files.

11.6. Release Documentation

Prepare release notes describing new features, fixes, known issues, and upgrade instructions.

11.7. Publishing Releases

Publish approved installers, documentation, and source code using the organization's release procedures.

11.8. Post-Release Verification

Validate installation on supported operating systems and confirm critical application functionality.

11.9. Release Checklist

Complete a formal checklist before marking the release as production ready.

12. Security and Data Protection

This section describes recommended practices for protecting application data, maintaining system integrity, and reducing operational risk throughout the lifecycle of RMA Tracker.

12.1. Security Overview

Security should be considered during installation, deployment, maintenance, and daily operation.

12.2. Database Protection

Restrict access to the SQLite database and include it in secure backup procedures.

12.3. File Permissions

Grant only the minimum required permissions to application, database, and backup directories.

12.4. Backup Security

Protect backup files from unauthorized access and verify their integrity regularly.

12.5. Secure Distribution

Distribute installation packages from trusted sources and verify package integrity.

12.6. Data Integrity

Validate critical operations and maintain reliable backups to preserve business data.

12.7. Future Authentication Features

Future releases may introduce user accounts, authentication, authorization, and audit logging.

12.8. Security Best Practices

Keep systems updated, limit administrative access, monitor backups, and document security procedures.

13.Best Practices

This section summarizes operational recommendations that improve system reliability, simplify maintenance, and encourage consistent administration of RMA Tracker.

13.1. Installation Best Practices

Verify prerequisites and perform installations using documented procedures.

13.2. Deployment Best Practices

Test deployments before production and document configuration changes.

13.3. Backup Strategy

Maintain automated backups and periodically test restoration procedures.

13.4. Upgrade Planning

Always create verified backups before applying application updates.

13.5. Maintenance Schedule

Follow documented daily, weekly, monthly, and annual maintenance activities.

13.6. Release Management

Maintain version history, release notes, and archived installation packages.

13.7. Disaster Recovery Planning

Document recovery procedures and periodically conduct recovery exercises.

13.8. Long-Term Support Recommendations

Review dependencies, update documentation, and plan future enhancements.

14. Appendices

This section contains supplemental reference material supporting installation, deployment, maintenance, troubleshooting, and future development of RMA Tracker.

14.1. Installation Directory Structure

Reference directory layout created during installation.

14.2. Application File Locations

Summary of executable, database, backup, and export locations.

14.3. Database File Reference

Overview of the SQLite database file and related resources.

14.4. Configuration File Reference

Description of configurable application files.

14.5. Build Commands

Frequently used Gradle build commands.

14.6. Gradle Commands

Reference for common Gradle tasks used during development.

14.7. jpackage Commands

Example packaging commands for supported platforms.

14.8. Release Checklist

Standard checklist for software releases.

14.9. Backup Verification Checklist

Checklist for validating backup integrity.

14.10. Recovery Checklist

Reference steps for disaster recovery.

14.11. Error Code Reference

Catalog of common errors and recommended resolutions.

14.12. Troubleshooting Decision Trees

Diagnostic workflows for common issues.

14.13. Glossary

Definitions of technical terms and abbreviations.

14.14. Revision History

Record of documentation revisions and publication history.